



examGenome CheatSheet on Algebra

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1	What is a whole number? Answer: A whole number is a number without fractions or decimals—starting from 0, 1, 2, 3, and so on. Explanation: Whole numbers are used for counting and never include negative numbers or parts like $\frac{1}{2}$ or 0.75.
2	What is the difference between even and odd numbers? Answer: Even numbers can be divided by 2 with no remainder (like 2, 4, 6). Odd numbers cannot (like 1, 3, 5). Explanation: You can check by looking at the last digit: if it's 0, 2, 4, 6, or 8, the number is even.
3	What is place value? Answer: Place value tells us the value of a digit based on its position in a number—like ones, tens, hundreds, etc. Explanation: In 345, the '3' is in the hundreds place, so it means 300—not just 3.
4	What is a prime number? Answer: A prime number is a number greater than 1 that has only two factors: 1 and itself. Explanation: Examples: 2, 3, 5, 7, 11. The number 1 is not prime because it only has one factor.
5	What are factors and multiples? Answer: Factors are numbers that divide evenly into another number. Multiples are the results of multiplying a number by whole numbers. Explanation: Factors of 6: 1, 2, 3, 6. Multiples of 6: 6, 12, 18, 24...
6	What is a square number? Answer: A square number is the result of multiplying a number by itself. Explanation: For example, $4 \times 4 = 16$, so 16 is a square number. It's called 'square' because it can form a perfect square shape with dots.
7	What is a number line used for? Answer: A number line shows numbers in order and helps with adding, subtracting, and comparing values. Explanation: Numbers increase as you go right and decrease as you go left. It also shows negative numbers to the left of zero.
8	What is rounding, and why do we do it? Answer: Rounding means making a number simpler but keeping its value close to the original. Explanation: We round to make numbers easier to work with—like saying 48 is 'about 50' when estimating.

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9	What are positive and negative numbers? Answer: Positive numbers are greater than zero. Negative numbers are less than zero and show values below zero (like temperature). Explanation: Zero is neither positive nor negative. Negative numbers have a minus sign (–) in front.
10	What is the rule for comparing two numbers? Answer: Start from the leftmost digit and compare place values. The number with the larger digit in the highest place is greater. Explanation: For example, 521 > 499 because 5 hundreds is more than 4 hundreds—even though 499 looks 'longer'.

